

Discover and Connect to Data in OneLake

Certification Track: DP-600 [Fabric Analytics Engineer Associate]

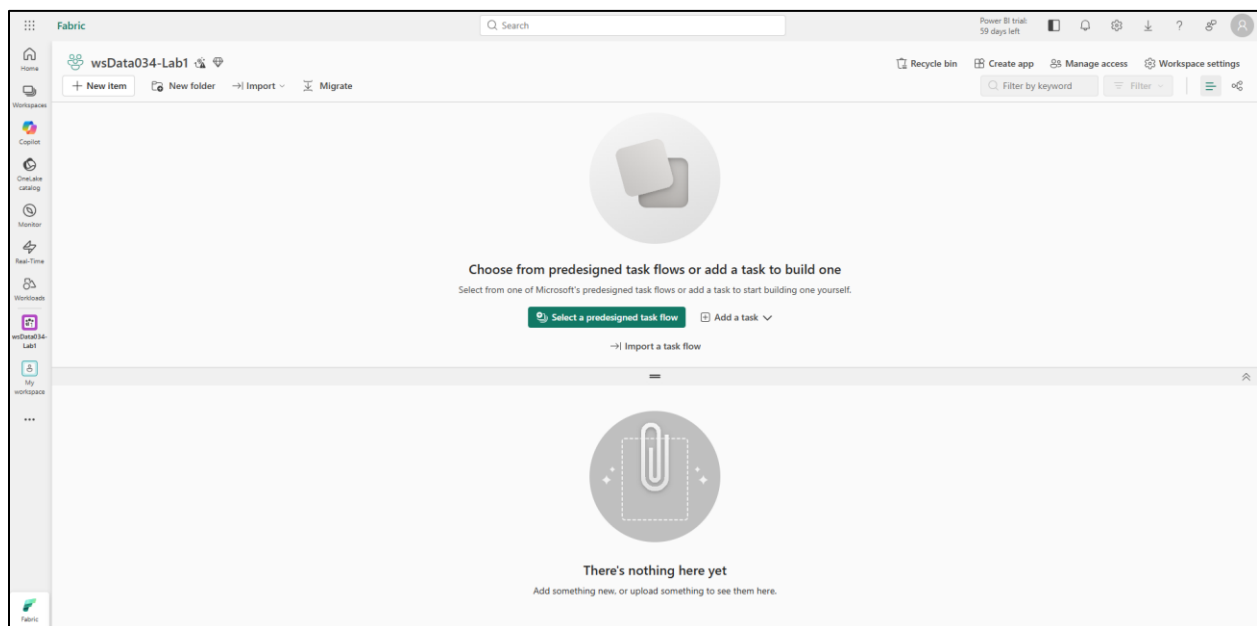
Data gets stored in different places across multiple teams within an organization. Instead of having to search through different workspaces or teams, Microsoft Fabric's OneLake Catalog allows you to search for the data you need, view metadata, understand, and govern all the data assets you have access to across your organization. The OneLake Catalog is a centralized place that holds one copy of all your organization's data.

In this lab you will build a sample Lakehouse with sales data, then use the OneLake catalog to locate it, create shortcuts, query through SQL endpoint, and explore semantic models.

1. Create Workspace

This is the starting point for any data project in Fabric. The first step is to create a workspace to hold and organize all your Fabric items.

- Open the web browser and go to the Fabric home page:
<https://app.fabric.microsoft.com/home?experience=fabric>
- Select Workspace from the left menu.
- Click +New workspace.
- Choose a Fabric and Power BI workspace type in the Advanced section. Select Fabric Trial and click Apply.
- Name your workspace and select Create. Your new workspace should be empty.

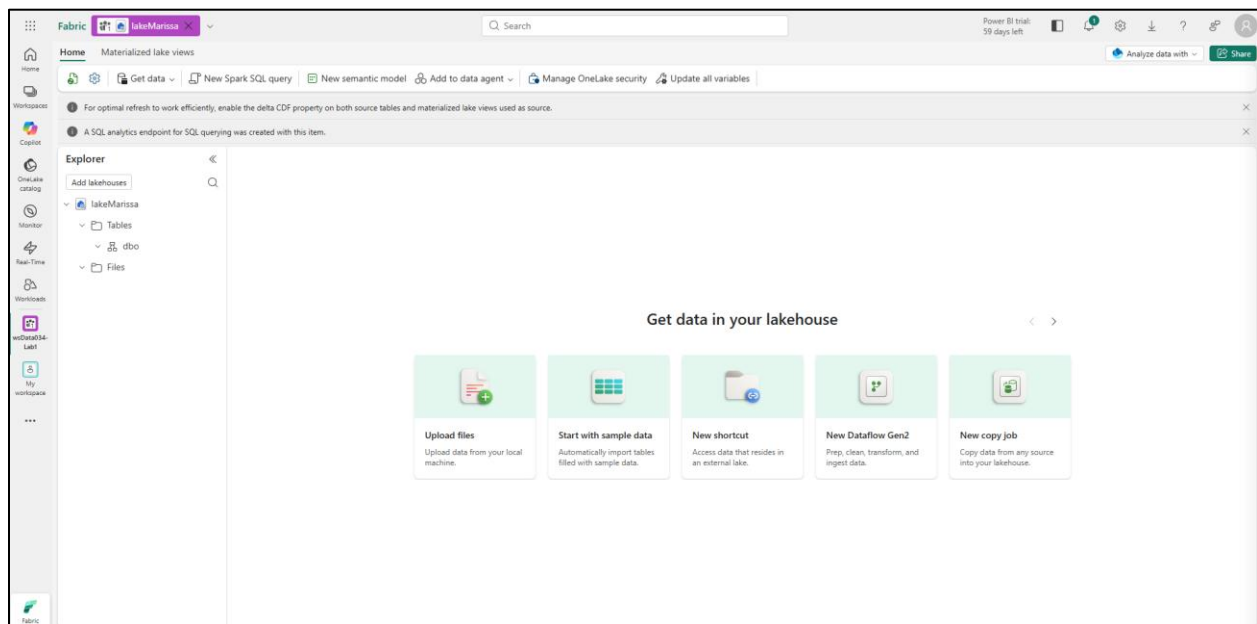


2. Create Lakehouse

A Lakehouse is an architecture which brings a datalake and a data warehouse together that provides flexibility and performance at the same time. It allows storing of raw, semi-structured and unstructured data and runs SQL on structured tables.

The second step is to add a Lakehouse to your workspace to store data. You will see that a new Lakehouse is created with empty Files and Tables folders.

- Inside your workspace select +New item.
- Choose Lakehouse.
- Enter a name for the Lakehouse.
- Click Create.

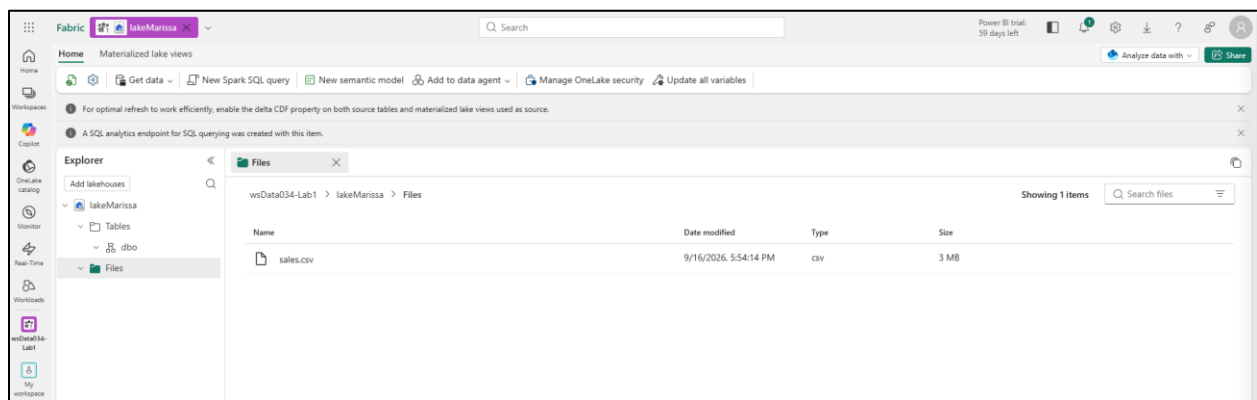
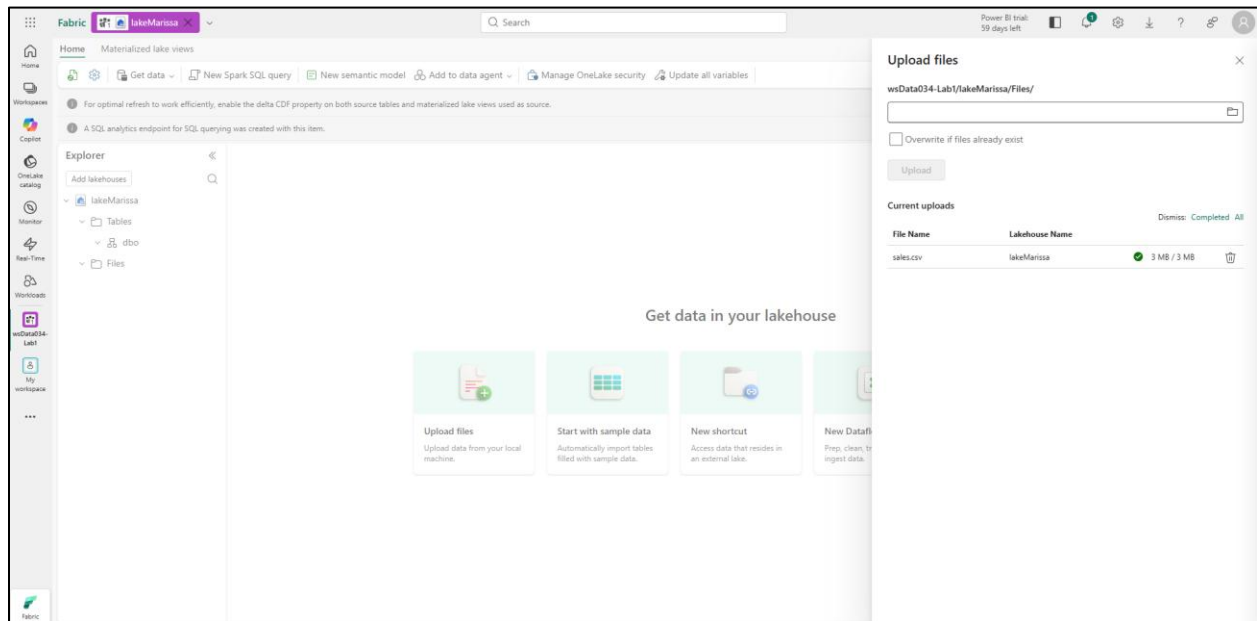


3. Load Sample Data into Lakehouse

Once your Lakehouse is set up, the next step is to populate it with sample sales data.

- Open browser in new tab and go to URL:
<https://raw.githubusercontent.com/MicrosoftLearning/dp-data/main/sales.csv>
- Download the CSV file and save it on your computer as sales.csv. Right-click anywhere on the page and select Save as to save it as a CSV file.
- ???Create a folder in your Lakehouse. Go to the Explorer pane-> Files folder, open the ... menu -> New subfolder. Name the folder data
- Upload the CSV. Go to the Explorer pane-> Files folder, open the ... menu on the Files folder-> Upload-> Upload files. Select sales.csv from your computer. Click Open then click Upload.

- Verify and preview. Open Files folder and confirm sales.csv is there. Select the file to preview its contents.

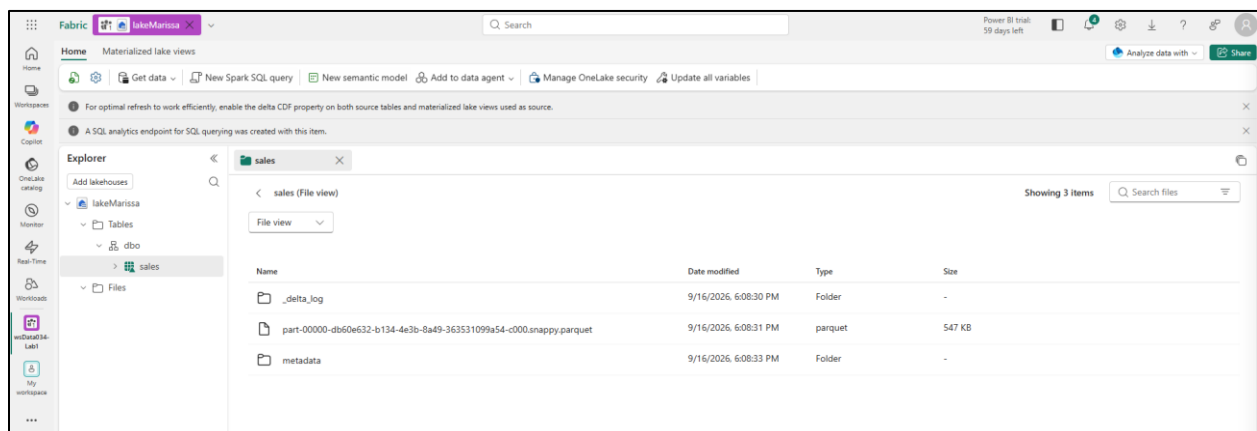
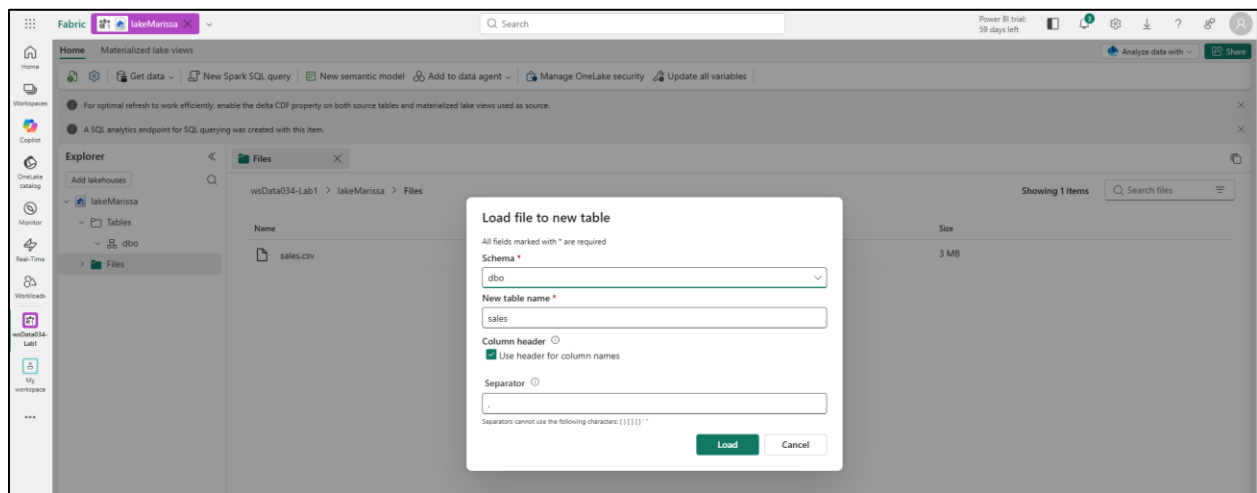
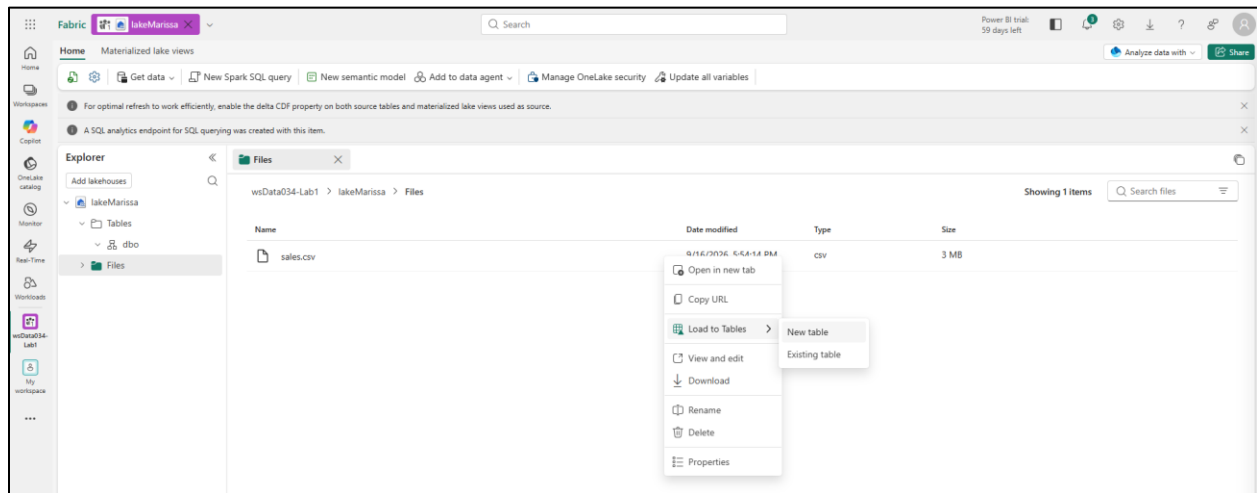


4. Load Data into Table

You can work with data files directly using Apache Spark. But to query the data with SQL, you need to load the data from the file into a table. Fabric will automatically create a Delta table which adds support for ACID transactions similar to a data warehouse.

- From Explorer pane, open the uploaded file sales.csv
- In the ... menu for the sales.csv file, choose Load to Tables-> New table.
- In Load to table dialog box, set the New Table name to sales and click Load.
- Confirm the load operation and wait for the table to be created and loaded.
- In the Explorer pane, check the Tables folder to see the new Delta table called sales.
- In the ... menu for the sales table, select File View to see the underlying files for this table.

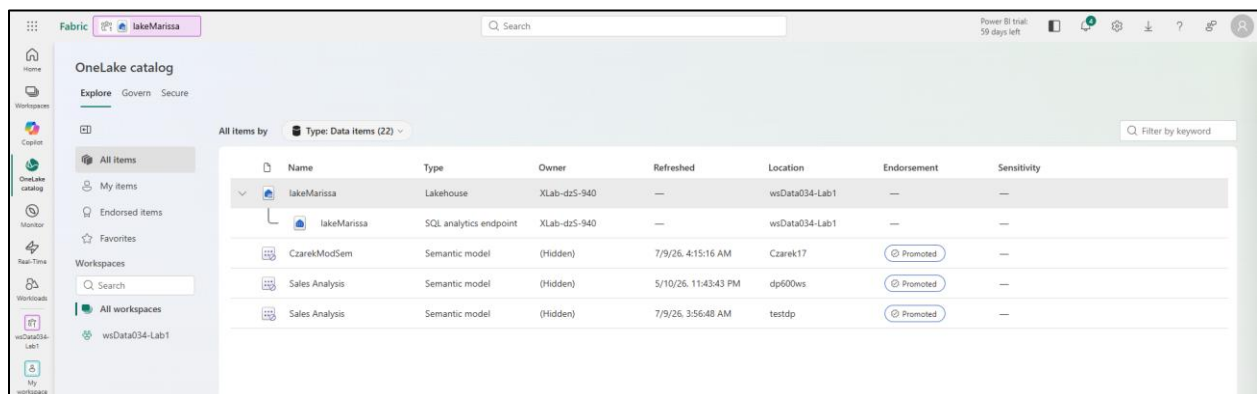
Notice the small triangle icon for the sales table. This means that all managed tables in the Lakehouse explorer are stored in Delta Lake format.



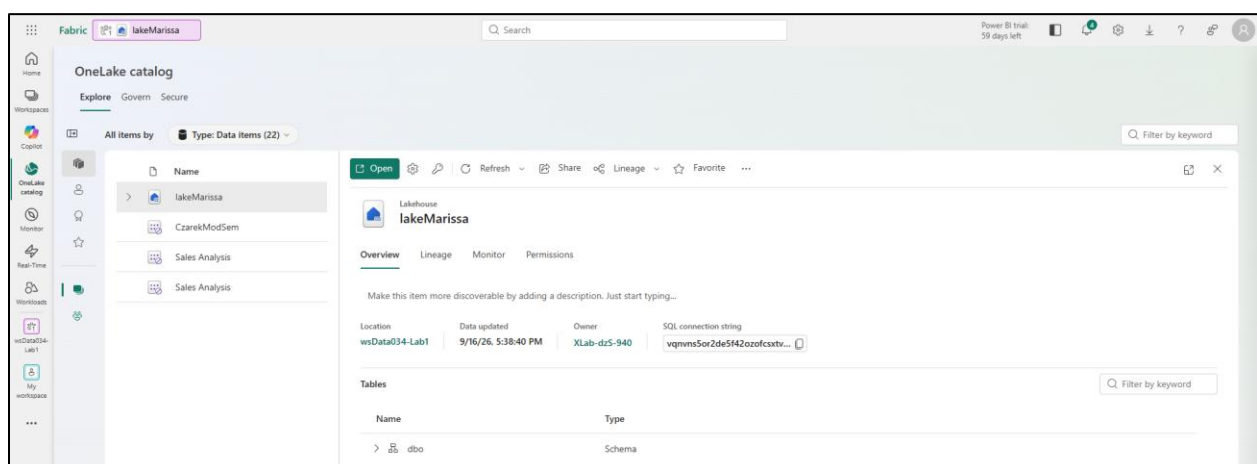
5. Browse the OneLake Catalog to Discover Data

Now that your Lakehouse contains data, you can find it using OneLake catalog. The OneLake catalog shows all data assets in one central place in your Fabric tenant.

- Select the Fabric icon at the top left of the page.
- In the left navigation pane, select the OneLake catalog icon. In the catalog, select the Explore tab. Make sure Data items is selected as the filter.
- Browse the list of data assets. You will see lakehouses, warehouses, semantic models, reports. The Type column shows what kind of data item it is. Find your Lakehouse.



- Select your lakehouse to view its details. The details pane show metadata information:
 - Location (workspace)
 - Data updated (last refresh time)
 - Owner (who created item)
 - SQL connection string (for tools like SSMS)



- Select Open to launch the Lakehouse explorer view

6. Explore Lakehouse Tables and Schema

The next step after locating a data asset in the catalog is to look at its structure to ensure the data aligns with the business needs and requirements. It is important that we have a clear understanding of the tables, columns and data types we're working with.

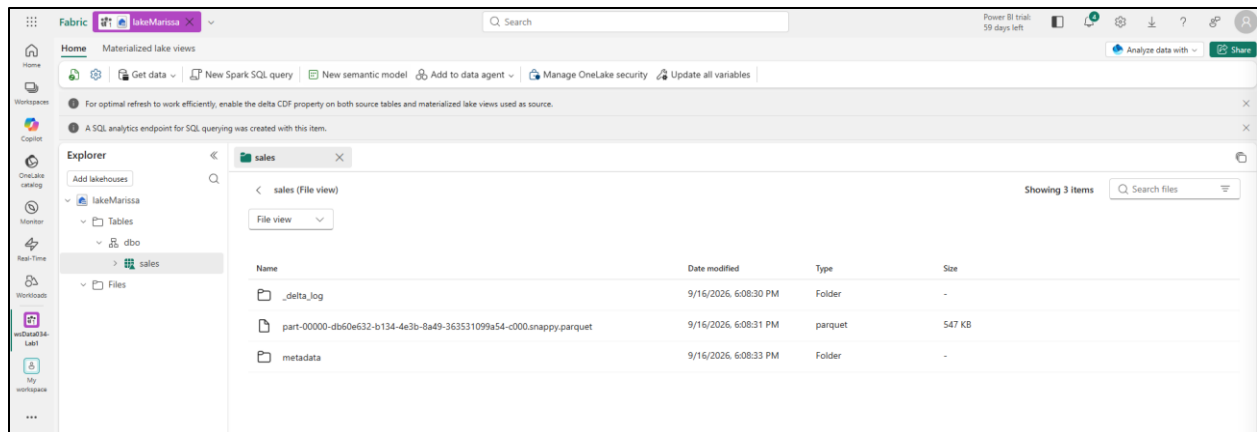
- In the catalog, select your lakehouse to open it.
- In the Explorer plane on the left, expand the tables folder to see the sales table.
- Select the sales table to show a data preview.
- In the previews, you can see columns like SalesOrderNumber, SalesOrderLineNumber, OrderDate, CustomerName, EmailAddress, Item, Quantity, UnitPrice, TaxAmount.
- Below the preview, the schema lists each column's data type.

	AKI SalesOrderNumber	103 SalesOrderLineNumber	OrderDate	AKI CustomerName	AKI EmailAddress	AKI Item	103 Quantity	12 UnitPrice	12 TaxAmount
1	SO43701	1	2019-07-01	Christy Zhu	christy12@adventure-wor...	Mountain-100 Silver, 44	1	3399.99	271.9992
2	SO43704	1	2019-07-01	Julio Ruiz	julio1@adventure-works...	Mountain-100 Black, 48	1	3374.99	269.9992
3	SO43705	1	2019-07-01	Curtis Lu	curtis9@adventure-works...	Mountain-100 Silver, 38	1	3399.99	271.9992
4	SO43700	1	2019-07-01	Ruben Prasad	ruben10@adventure-works...	Road-450 Black, 62	1	699.0982	55.9279
5	SO43703	1	2019-07-01	Albert Alvarez	albert17@adventure-works...	Road-150 Red, 62	1	3578.27	286.2616
6	SO43697	1	2019-07-01	Cole Watson	cole1@adventure-works...	Road-150 Red, 62	1	3578.27	286.2616
7	SO43699	1	2019-07-01	Sydney Wright	sydney61@adventure-works...	Mountain-100 Silver, 44	1	3399.99	271.9992
8	SO43702	1	2019-07-01	Colin Anand	colin43@adventure-works...	Road-150 Red, 44	1	3578.27	286.2616
9	SO43698	1	2019-07-01	Rachael Martinez	rachael16@adventure-works...	Mountain-100 Silver, 44	1	3399.99	271.9992
10	SO43707	1	2019-07-02	Emma Brown	emma3@adventure-works...	Road-150 Red, 48	1	3578.27	286.2616
11	SO43711	1	2019-07-02	Courtney Edwards	courtney1@adventure-works...	Road-150 Red, 56	1	3578.27	286.2616
12	SO43706	1	2019-07-02	Edward Brown	edward26@adventure-works...	Road-150 Red, 48	1	3578.27	286.2616
13	SO43708	1	2019-07-02	Brad Deng	brad2@adventure-works...	Road-450 Red, 52	1	699.0982	55.9279
14	SO43709	1	2019-07-02	Martha Xu	martha12@adventure-works...	Road-150 Red, 52	1	3578.27	286.2616
15	SO43710	1	2019-07-02	Katrina Raji	katrina30@adventure-works...	Road-150 Red, 56	1	3578.27	286.2616
16	SO43712	1	2019-07-02	Abigail Henderson	abigail73@adventure-works...	Road-150 Red, 44	1	3578.27	286.2616
17	SO43720	1	2019-07-03	Melanie Sanchez	melanie47@adventure-works...	Road-150 Red, 44	1	3578.27	286.2616
18	SO43721	1	2019-07-03	Louis Xie	louis20@adventure-works...	Road-150 Red, 62	1	3578.27	286.2616
19	SO43714	1	2019-07-03	Latacha Alonso	latacha8@adventure-works...	Road-150 Red, 44	1	3578.27	286.2616
20	SO43715	1	2019-07-03	Warren Jai	warren42@adventure-works...	Road-150 Red, 56	1	3578.27	286.2616
21	SO43716	1	2019-07-03	Jonathan Gutierrez	jonathan3@adventure-works...	Road-150 Red, 44	1	3578.27	286.2616

- Switch to File view to display the underlying files for the sales table.

The sales table is a Delta Lake table in Fabric which is just a folder in OneLake that contains two kinds of files:

- 1) Parquet files that store the data in the sales table. Parquet files are column-based files designed for efficiency.
- 2) A transaction log (_delta_log folder) that tracks every change to the table.

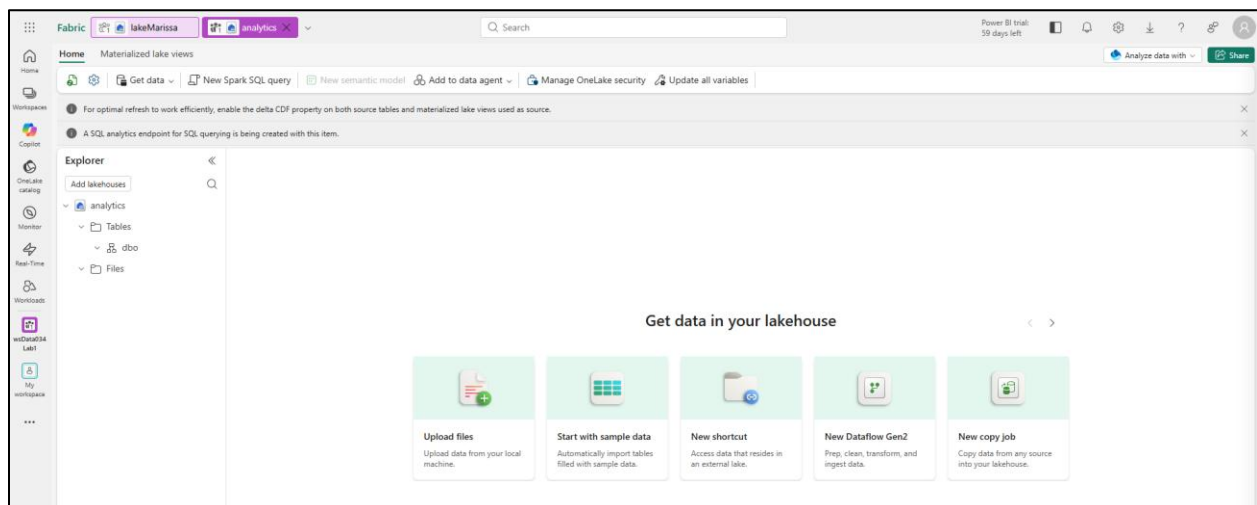


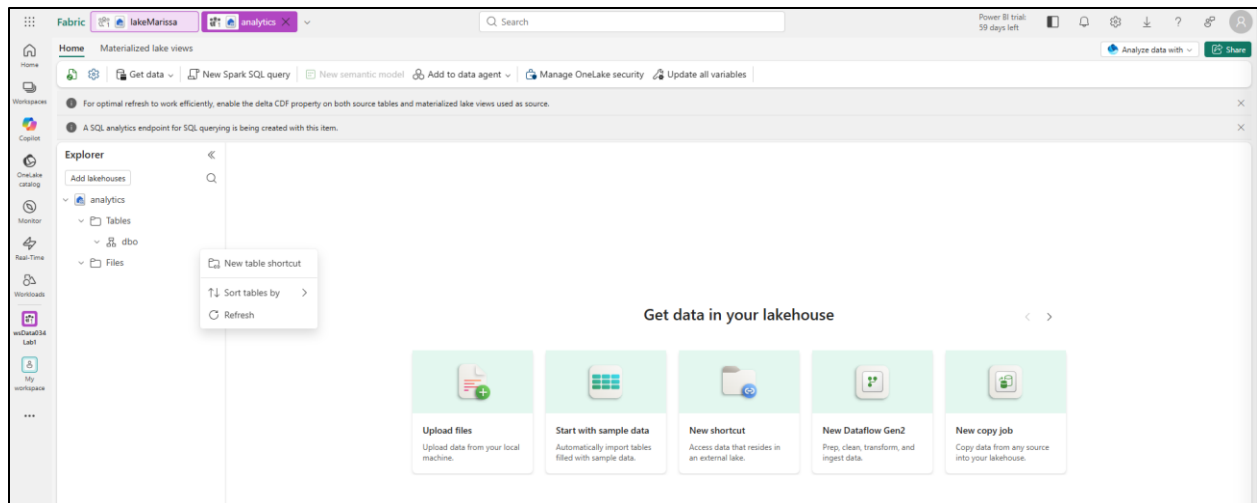
7. Create a Shortcut to access Data from Another Workspace

The purpose of a shortcut is to keep one copy of data. A shortcut is a pointer to data stored somewhere else. Fabric lets you access the data using a live connection to its original source without copying.

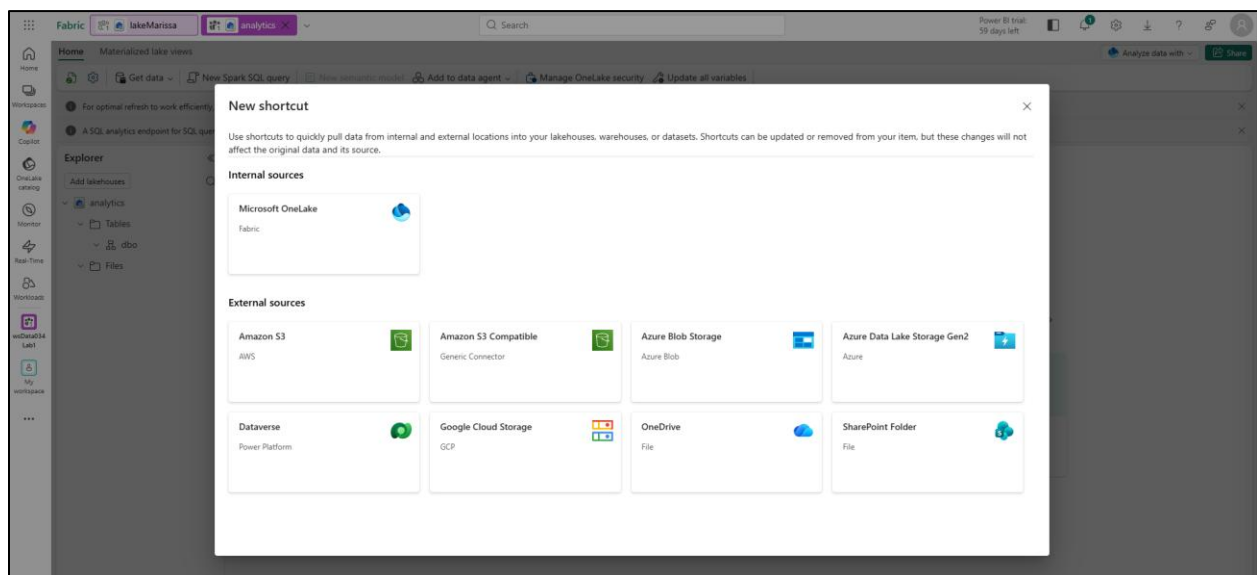
This step describes how to create a shortcut from your original lakehouse to a second lakehouse.

- Select your workspace name icon from the left column.
- Create a second lakehouse by clicking +New item> Lakehouse. Name the lakehouse analytics.
- When the lakehouse opens, in the Explorer pane, expand Tables and select ... menu for the dbo folder and select New table shortcut.

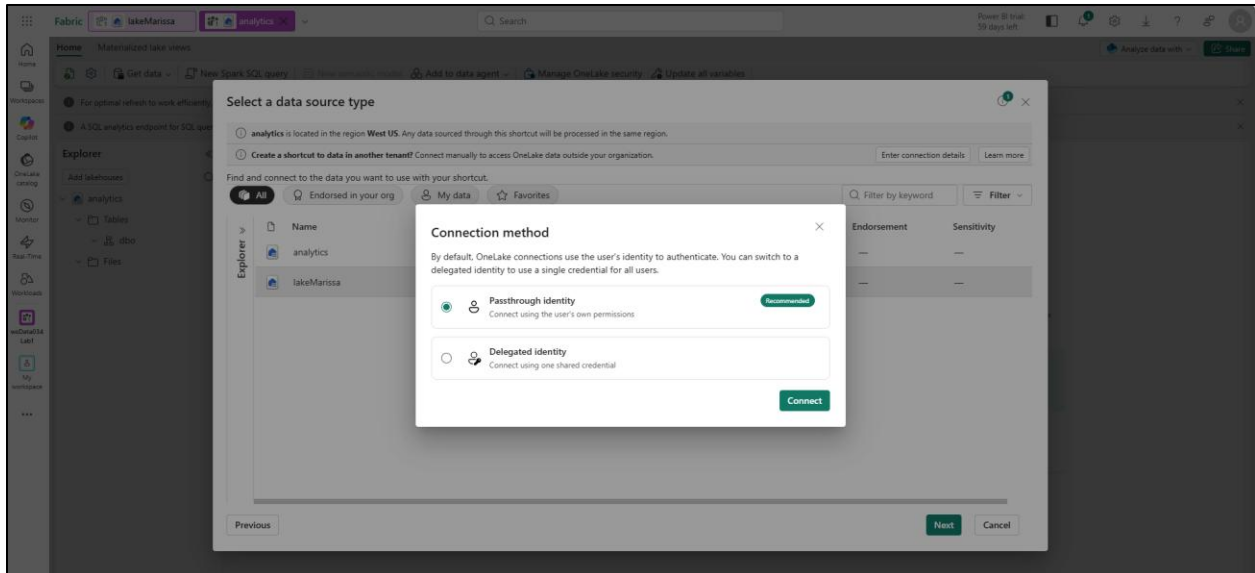




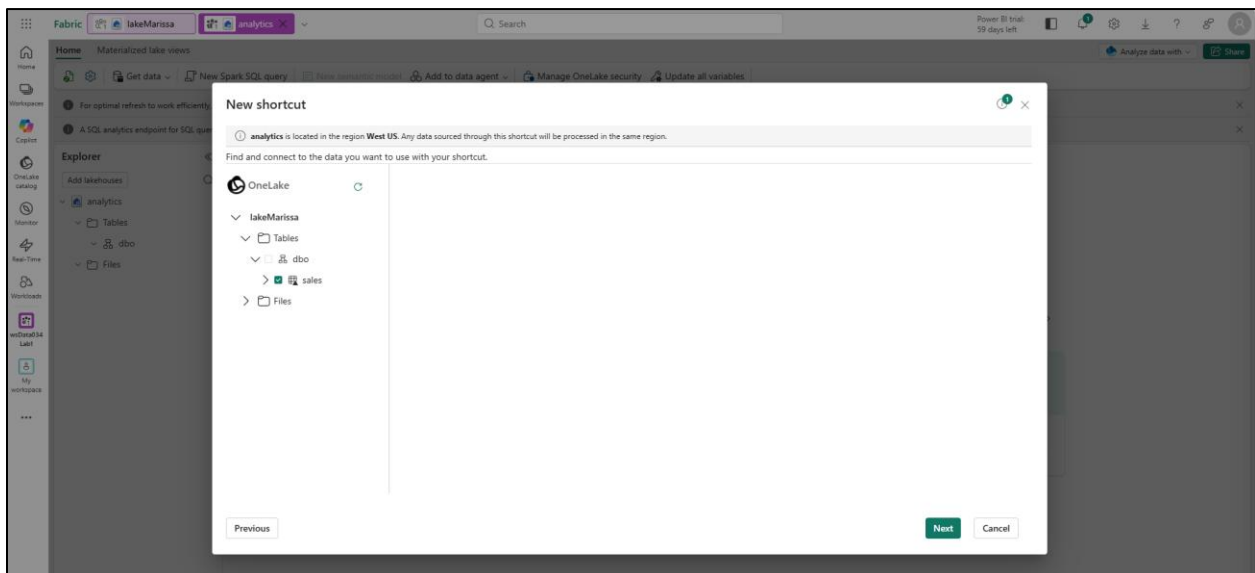
- The New shortcut dialog window opens. Click on Microsoft OneLake as the shortcut source.



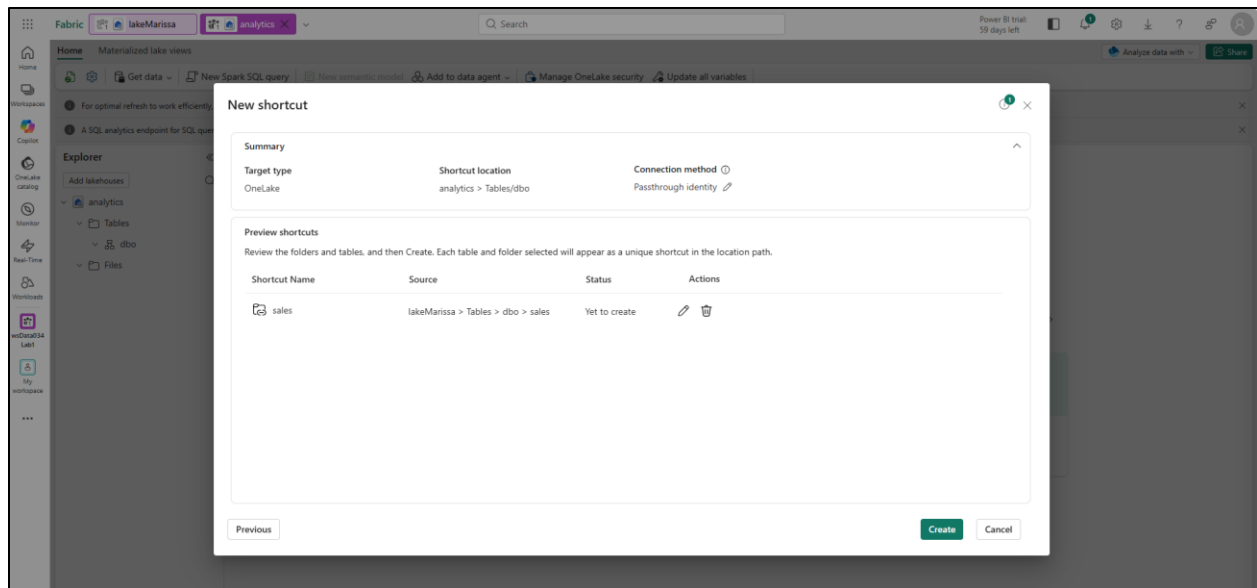
- In the workspace list, select the workspace containing your original lakehouse.
- Select your original lakehouse.
- Select Passthrough identity as the Connection method.



- Select Tables table, and then select Next.



- Review the shortcut details before selecting Create.
 - Target type: Which shortcut type, OneLake in this case
 - Shortcut location: The lakehouse you want the shortcut in
 - Shortcut name: Defaults to sales
 - Source: The lakehouse where the data exists



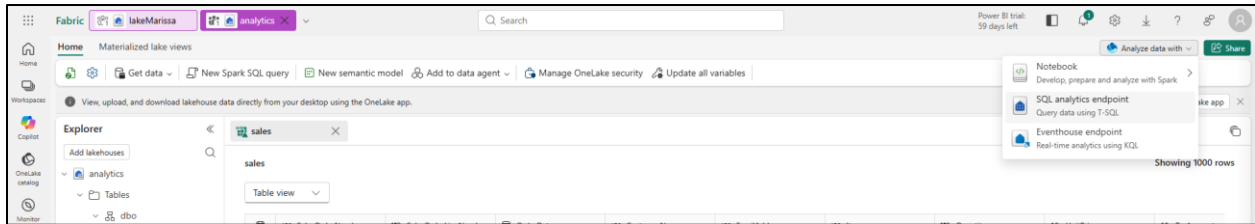
- From the analytics lakehouse in the Explorer pane, expand the Tables folder. The sales table has a shortcut icon (a link overlay). This means that this sales table is just referencing or pointing to the data in your original lakehouse.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	SO43701	1	2019-07-01	Christy Zhu	christy12@adventure-wor...	Mountain-100 Silver, 44	1	3399.99	271.9992							
2	SO43704	1	2019-07-01	Julio Ruiz	julio1@adventure-works...	Mountain-100 Black, 48	1	3374.99	269.9992							
3	SO43705	1	2019-07-01	Curtis Lu	curtis9@adventure-works...	Mountain-100 Silver, 38	1	3399.99	271.9992							
4	SO43700	1	2019-07-01	Ruben Prasad	ruben10@adventure-works...	Road-650 Black, 62	1	699.0982	55.9279							
5	SO43703	1	2019-07-01	Albert Alvarez	albert7@adventure-works...	Road-150 Red, 62	1	3578.27	286.2616							
6	SO43687	1	2019-07-01	Cole Watson	cole1@adventure-works...	Road-150 Red, 62	1	3578.27	286.2616							
7	SO43699	1	2019-07-01	Sydney Wright	sydney61@adventure-wor...	Mountain-100 Silver, 44	1	3399.99	271.9992							
8	SO43702	1	2019-07-01	Colin Anand	colin45@adventure-works...	Road-150 Red, 44	1	3578.27	286.2616							
9	SO43698	1	2019-07-01	Rachael Martinez	rachael16@adventure-wor...	Mountain-100 Silver, 44	1	3399.99	271.9992							
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11	SO43711	1	2019-07-02	Courtney Edwards	courtney1@adventure-wor...	Road-150 Red, 56	1	3578.27	286.2616							
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15	SO43710	1	2019-07-02	Katrina Raji	katrina20@adventure-wor...	Road-150 Red, 56	1	3578.27	286.2616							
16	SO43712	1	2019-07-02	Abigail Henderson	abigail77@adventure-wor...	Road-150 Red, 44	1	3578.27	286.2616							

8. Query Data using the SQL Analytics Endpoint

When a lakehouse is created, Fabric automatically creates an SQL endpoint that enables you to query tables using SQL to easily explore data. This is a fast, serverless, read-only interface to run standard T-SQL SELECT statements without using Spark.

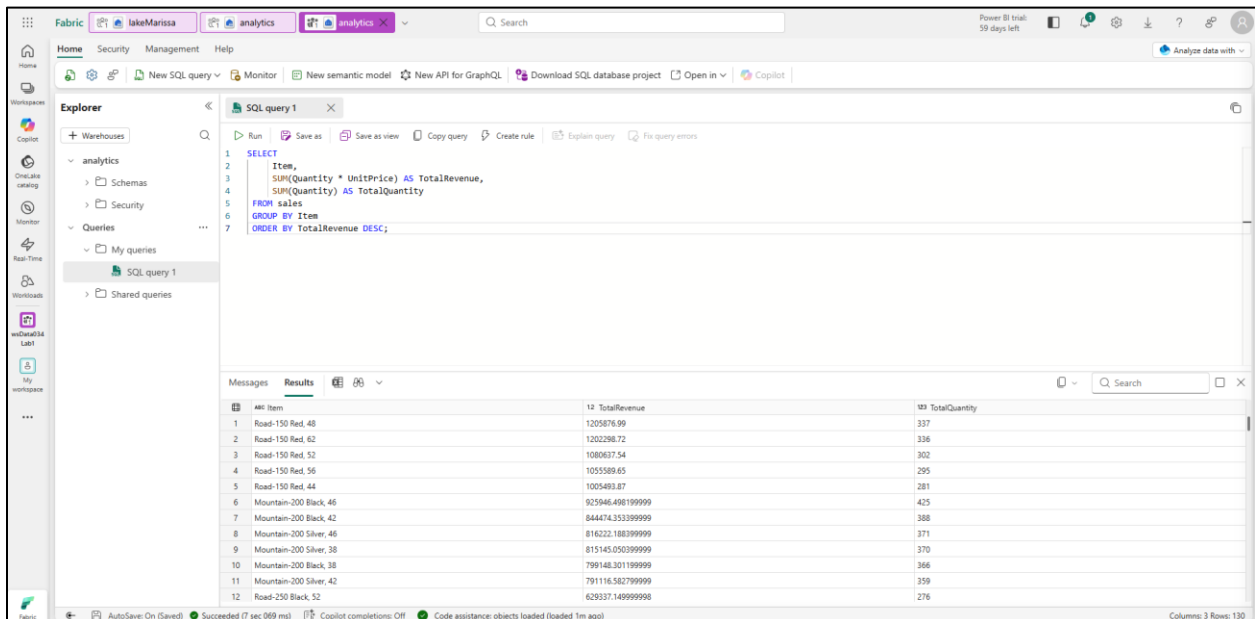
- The top right of the page shows a drop down “Analyze data with”. Click the drop down arrow and select SQL analytics endpoint.



- In the toolbar, select New SQL query to open the query editor and enter the following T-SQL query calculates total revenue and total quantity sold for each item:

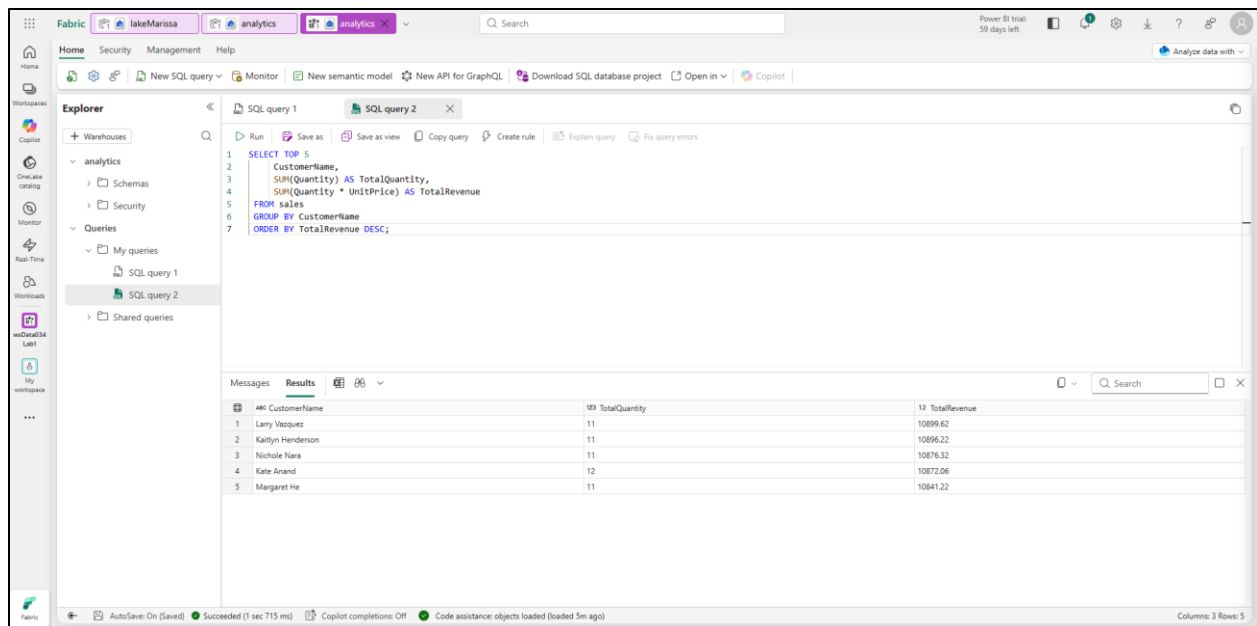
```
SELECT
    Item,
    SUM(Quantity * UnitPrice) AS TotalRevenue,
    SUM(Quantity) AS TotalQuantity
FROM sales
GROUP BY Item
ORDER BY TotalRevenue DESC;
```

- Click Run to execute the query and the results will appear below showing revenue by item.



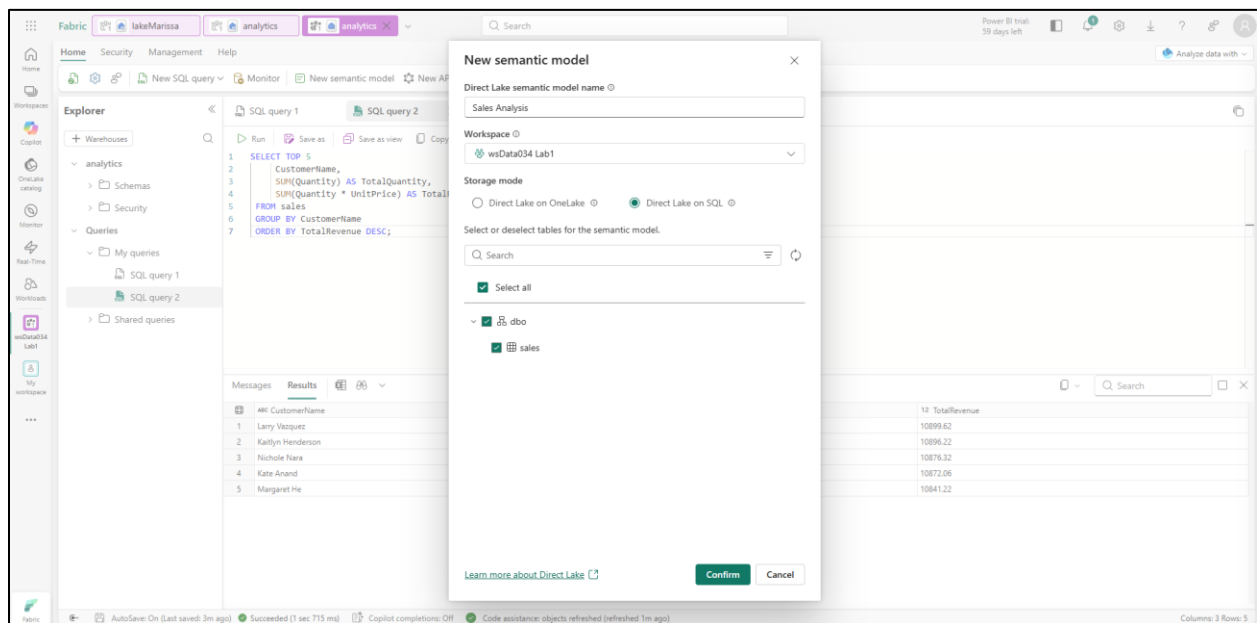
- Write a second query to find the top customers by revenue:

```
SELECT TOP 5
    CustomerName,
    SUM(Quantity) AS TotalQuantity,
    SUM(Quantity * UnitPrice) AS TotalRevenue
FROM sales
GROUP BY CustomerName
ORDER BY TotalRevenue DESC;
```



9. Create and Explore a Semantic Model

Semantic models provide a business-friendly layer on top of data, including predefined relationships, measures, and calculations for Power BI reports. In this task, you create a semantic model from the lakehouse and explore it using the Explore this data feature.



Fabric lakeMarissa analytics analytics

wsData034 Lab1

Choose from predefined task flows or add a task to build one

Select from one of Microsoft's predefined task flows or add a task to start building one yourself.

Select a predefined task flow Add a task

Import a task flow

Name	Status	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
analytics		Lakehouse	—	XLab-vNG-915	—	—	—	—	
analytics		SQL analytics en...	—	XLab-vNG-915	—	—	—	—	
lakeMarissa		Lakehouse	—	XLab-vNG-915	—	—	—	—	
lakeMarissa		SQL analytics en...	—	XLab-vNG-915	—	—	—	—	
Sales Analysis		Semantic model	—	wsData034 Lab1	9/17/2026, 11:35:43 ...	N/A	—	—	

Fabric lakeMarissa analytics analytics

wsData034 Lab1

Explore Sales Analysis (preview)

Home

Open in new tab Share

Add filter Remove filters

Matrix Sort Drill Totals New visual calculation

Item	Sum of Quantity
Water Bottle - 30 oz.	2097
Patch Kit/8 Patches	1621
Mountain Tire Tube	1581
David Tire Tube	1515
Total	32718

Visual

Sum of Quantity by Item

Water Bottle - 30 oz.

Patch Kit/8 Patches

0 500 1000 1500 2000

Data

Choose data

Search

sales

CustomerName

EmailAddress

Item

OrderDate

Quantity

SalesOrderLine...

SalesOrderNumber

TaxAmount

UnitPrice

Rearrange data

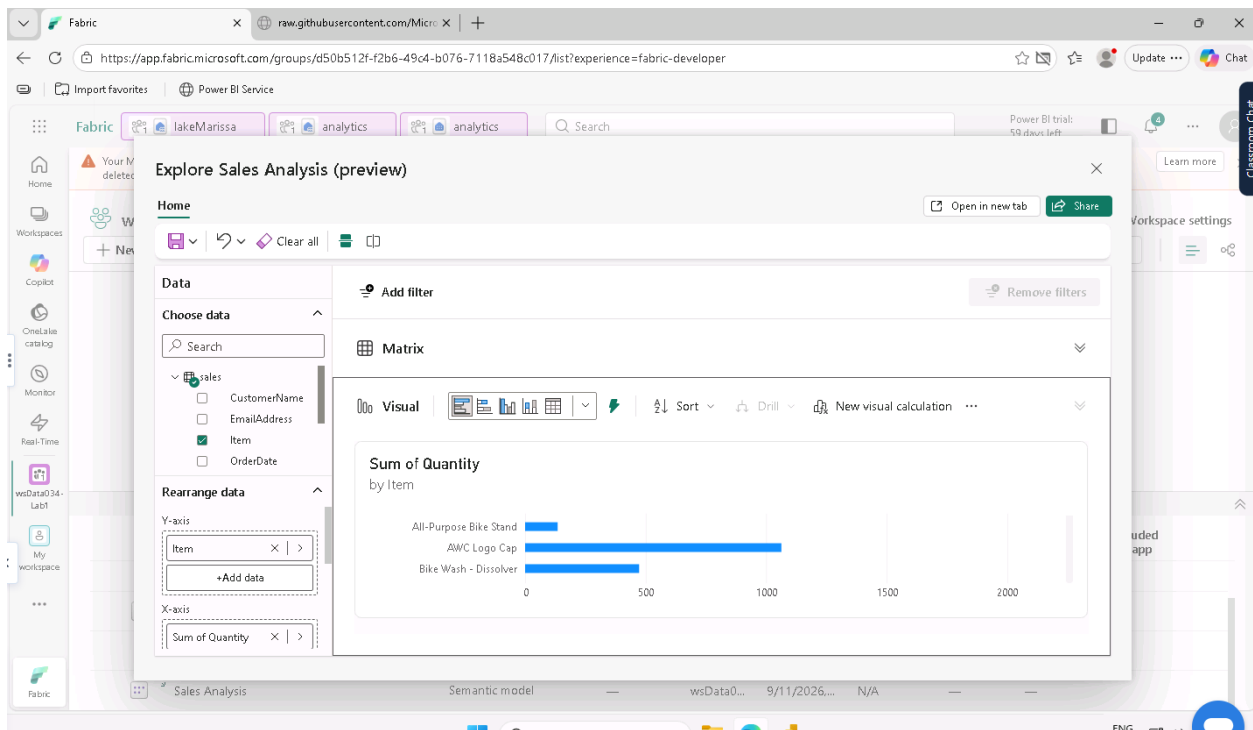
Rows

Item

Columns

Sum of Quantity

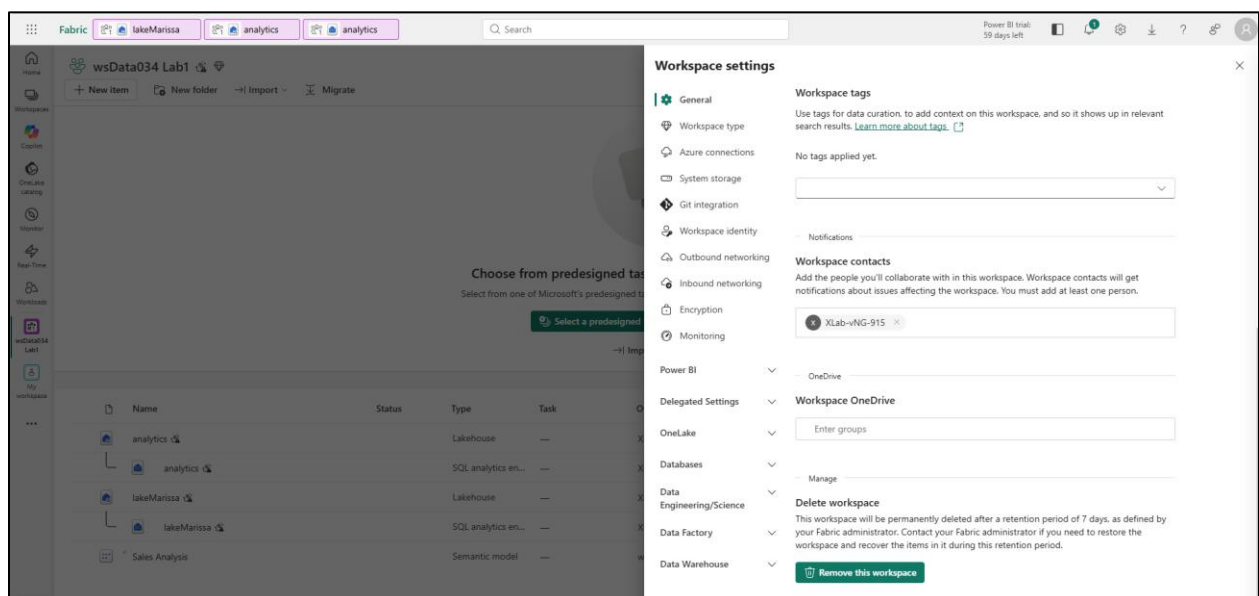
Values



10. Clean up Resources

It is important to delete the resources after completing the lab to stop unnecessary capacity consumption which incur costs.

- Select the icon for your workspace on the left to view all items in the workspace.
- In the toolbar, select Workspace settings.
- In the General section, select Remove this workspace.



11. Reflection

The most important thing I learned from this lab is that all data in Microsoft Fabric is stored in a single, unified place called OneLake. OneLake serves as the foundational storage layer that every Fabric workload reads from and writes to. I also learned that the OneLake catalog plays a key role by making this data easy to discover, understand, and access across the organization. This ensures that users can quickly search, connect to and work with the right data without needing to know where it physically lives.

You start the lab by creating a workspace, adding a lakehouse to it and then loading sample data in the lakehouse to generate a table. Once that's done, you can browse this data along with all other data assets in the OneLake Catalog. Shortcuts allow you to access data stored in other locations without copying. SQL endpoint gives you an instant, read-only interface over your lakehouse tables. Finally, a semantic model can be created from a lakehouse to provide a clean, business-friendly layer for reporting. The key takeaway is that OneLake provides one shared copy of data that the entire organization can use for cross-team collaboration, ensuring everyone works from the same trusted source.